

AI-Enhanced Metaheuristics Control for Project Scheduling Using Neural Networks

ABSTRACT

Every project is made up of many small tasks that depend on each other and share limited resources such as workers, machines, or money, and finding the best order and timing to complete all tasks as quickly as possible is known as Project Scheduling, a problem that becomes extremely difficult to solve exactly as the project grows larger. To handle this, researchers use smart search techniques called metaheuristic algorithms, such as Genetic Algorithm (GA), Simulated Annealing (SA), Bee Colony Optimization (BCO), and simple Priority Rule based Heuristics (PRH), but since no single algorithm works best for every project, a recent study trained a small Artificial Neural Network that looks at three simple numbers describing a project's structure and instantly recommends which of these four algorithms is likely to give the best schedule, avoiding the need to run every algorithm and compare results. This project builds on that idea and improves it in eight ways, with the most significant contribution being the addition of two brand-new algorithms that the original study did not include: Tabu Search (TS), which remembers recently tried solutions and avoids repeating them so the search does not get stuck going back and forth between the same options, and Particle Swarm Optimization (PSO), inspired by how a flock of birds searches for food together, where each candidate schedule adjusts itself based on its own best result and the best result found by the whole group to quickly converge on strong schedules; together these expand the neural network's choices from four algorithms to six, giving it a wider and more capable set of strategies to recommend from. The remaining improvements include using richer project features for better accuracy, predicting the actual expected schedule duration rather than only naming an algorithm, providing a confidence score with each recommendation, correcting a data imbalance issue so rarely-best algorithms are still recognized reliably, testing the system on newly generated unseen projects to check real-world generalization, using a Graph Neural Network to read a project's task structure directly, and a self-adjusting controller that can switch algorithms mid-search if the current one stalls, together producing a faster, more accurate, and adaptable assistant that saves computing effort while giving schedules as good as or better than the original system.

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